

Breezy the Calm Monster: Soft Toy Design Combined with Pervasive Technology to Teach Deep Breathing

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ABSTRACT

Anger dysregulation can lead to aggression and peer rejection in early childhood. Anger episodes and temper tantrums frequently escalate, leaving parents and young children needing effective emotion regulation interventions. This research investigates how designing a playful interaction with ubiquitous technology can help parents teach young children emotion regulation strategies such as deep breathing. Breezy is a soft toy with an app and a storybook that helps parents and engages them in teaching deep breathing as an anger regulation strategy. The soft toy has sensory and physical features and can be held as a puppet, which helps young children to connect with the toy. The digital app addresses facial expressions and physiological components of emotions through sensory and interactive features that encourage identifying emotions. The storybook facilitates a familiar context for parents' engagement through playful learning. The interactive system provides parents with opportunities to teach their young children to identify and label anger and practice and model age-appropriate breathing exercises. This research aims to inform the design of future educational and playful interactions for emotional competence skills, as well as to broaden their application and promote their use in homes.

CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile computing; Ubiquitous and mobile devices; Smartphones.

KEYWORDS

Emotion regulation, Parent-child interaction, research through design, Soft toys

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1 INTRODUCTION

Young children can experience difficulties developing emotional competence (EC) skills. Reports suggest that 8% of New Zealand children between the ages of three and 14 years old (57,000 children) experience social, emotional, and behavioural problems [1]. Children between three and four years of age had a higher rate of experiencing such complications at 10.2 % [1]. Internationally, 14% (1 in 7) of 10- to 19-year-olds worldwide have mental health conditions, many of whom have not received formal diagnoses or treatment [2]. Early childhood may be a particularly beneficial time to intervene, as interventions can reduce behavioural difficulties while increasing emotional competencies [3]. Moreover, anger episodes and tantrums peak at this age [4, 5], leaving parents and their children needing effective emotion regulation (ER) interventions.

Digital technologies and parents' involvement can help teach young children ER strategies [6, 7]. However, most technology-based tools and interventions are explicitly created for adult users compared to children, and when they do focus on children, they often address neurodivergent children. Using everyday playful objects that young children (two to three years old) enjoy and are motivated to engage with offers the potential to deliver a meaningful form of parent-directed playful learning. While many existing interventions, such as educational programs, toys, and apps, are likely to be beneficial to the development of children's EC skills, there is an opportunity to design a smart toy that assists parents in teaching ER skills for home-based play interactions. This approach may encourage ER learning, reducing emotional and behavioural difficulties while increasing emotional competencies.

2 IMPLEMENTATION

2.1 Design Approach

This paper presents the design of a playful system to help parents teach young children anger regulation strategies through a soft toy, storybook, and interactive app. While the design concepts provide

playful and tangible interaction with digital integration, they introduce an approach for a novel toy design concept by engaging parents in modelling the ER strategies.

In the first phase of the research, we interviewed experts in the field and parents based on a human-centred design (HCD) methodology to understand the needs and requirements of children and their parents and develop a set of design criteria [8] that guided the creation of many design concepts. In the second phase, an iterative and expansive [9] research through design (RTD) process was used to develop many design concepts. Initial concepts were assessed using a focus group with designers and parents and a weighted matrix model [10, 11]. In the third phase, a serial [9] iterative RTD study was conducted to refine the design concept. The prototypes were evaluated accordingly until a final design concept was formed for testing. The prototype was initially tested by experts and parents with their children.

2.2 Technology-Use and Implementation

The aim of this project is to help parents engage in play, so that they can teach their young children (two-three years old) identify and label anger and practise a deep breathing exercise. We constructed design criteria based on several physical and cognitive challenges to achieve this aim: 1) the design should describe physiological changes when experiencing anger and use facial expressions to identify the emotion; 2) the design should demonstrate deep breathing exercise steps simply; 3) the design should enable parents to model the strategy; 4) the design should be easy to set up and play with, reducing obstacles to implementing the learning; 5) the design aesthetics should appropriately reflect the expected play for children aged two to three years old; and 6) the designed system should also have a safe, familiar, and suitable form for domestic use. Considering these requirements and after assessing hundreds of concepts, we focused on a pervasive technology as a tangible user interface (TUI): a smartphone digital application that shows facial and physiological expression visually and audibly for each emotion. Our TUI utilises sensory and interactive features to explore emotions while applying the benefits and qualities of the tangible soft toy to guide the playful interaction.

Based on the feedback from the two focus group studies, the literature review, and existing toys, four considerations were assessed in the final toy prototype.

- Digital technology in the form of a TUI was chosen to develop the design concept and could help support parents in teaching. From the experts' perspective, the TUI, through an application on a smartphone, gave the physical stuffed toy a voice, personality, and explicit facial expressions, removing any symbolic elements. Screen-based devices like smartphones and tablets have allowed children as young as two-years-old to engage with digital content [12, 13].
- Soft toys can be used as physical and transitional objects to support children's cognitive, social, and emotional growth. Soft toys create a familiar, comfortable, and emotionally engaging context for children, allowing for modelling, offering feedback, and representing emotions. Monstrous forms were chosen to make a gender-neutral play object more appealing and inclusive.

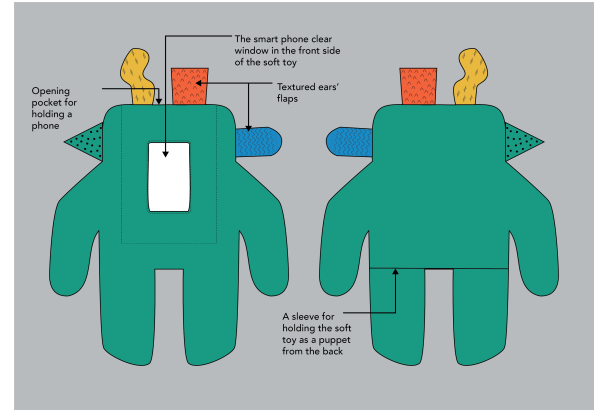


Figure 1: Front and rear-view illustration of the soft toy

- The deep breathing exercises should be accessible and functional for young children, and parents should be able to model them using the toy. One of the deep breathing exercises is belly breathing, which involves lying on the floor with a teddy bear or other safe object on the belly and breathing in and out. Teaching children how to breathe deeply using their diaphragm was simple and inexpensive because it could be done through explanations, modelling, and imitation [14, 15]. We incorporated belly breathing exercises into the design. Experts commonly used them to teach children how to practice mindfulness.
- We included a storybook, a common medium, to help involve parents in the experience. Interactive storytelling with physical objects can also help children develop their imagination. The story's context included a deep breathing exercise.

2.3 The System Components

In the following section, we present the final design of the playful interaction that combines three different components of physical and digital play.

The design of the system consists of three components: (A) a physical soft toy, (B) a storybook, and (C) an interactive application that can be inserted into the soft toy acting as the face of the toy.

We argue that while pervasive technology through an app was required to provide interactive experiences that respond to the situation (for instance, the toy can show different emotions through facial expressions), the physical toy allowed children to develop a relationship with the system without being constantly exposed to technology. The storybook enabled parents to participate in the experience through a familiar medium. The interactive system provides a playful interaction that parents can implement in their everyday lives to teach their children how to identify and manage anger through deep breathing.

2.3.1 The Soft Toy. The physical soft toy features a green monster with textured ears and a front pocket with a transparent opening to hold a smartphone (Figure 1). A sleeve from the back fits a hand for holding the soft toy as a puppet (Figure 2). The final design of the



Figure 2: Accessible back pocket for simple manipulation of the toy

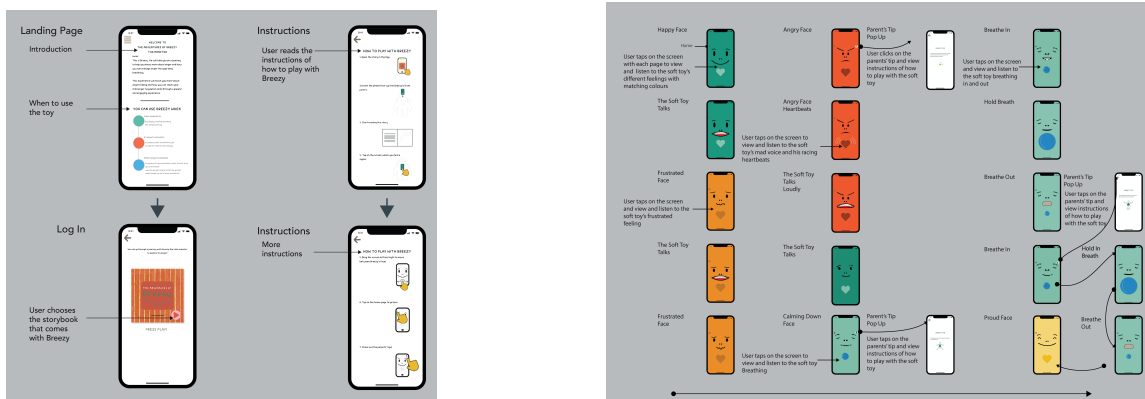


Figure 3: The interface design (left) and an illustration show the different facial expressions of Breezy in the app (right).

full-size soft toy was used to show the different feelings described on the application's interface and inside the toy's body.

2.3.2 The Interactive Digital Application. The smartphone is placed in the toy's see-through front pocket to serve as the monster's face (Figure 3). The interface focuses on facial expressions to help children label the soft toy's emotions through auditory feedback and presented in colours: yellow for happiness, orange for frustration, red for anger, and green for calm (Figure 3). The interface also provides physical expressions of the emotion to help children recognise its signs. For instance, each facial expression triggered a label of the emotion in the form of a sound (e.g., "I am mad!") and physiological signs (e.g., a sound of rapid heartbeats) to encourage interaction with the toy, provide auditory support for the expression, and offer a voice for the toy. Facial expressions and auditory feedback will be matched with background-similar music to create an immersive experience. For example, when a yellow

facial expression is displayed, positive and uplifting music will be provided to match the visuals with the narrative.

2.3.3 Storytelling. The storybook of the adventures of Breezy guides parents in teaching belly breathing exercises as a strategy to manage anger (Figure 4). The story contains tips to guide parents. Three activity pages offer a recap and collaborative tasks for engagement.

3 TESTING FINDINGS

According to our preliminary usability testing findings, the interactive toy system can assist parents in teaching anger labelling and the deep breathing exercise to their young children. Participants responded positively to the toy prototype, and children easily adapted to our prototype soft toy, app integration, and storytelling approach, with two out of four children completing the belly breathing exercise. According to parents and expert participants, testing revealed that the interactive system's hybrid physical (soft toy and story) and

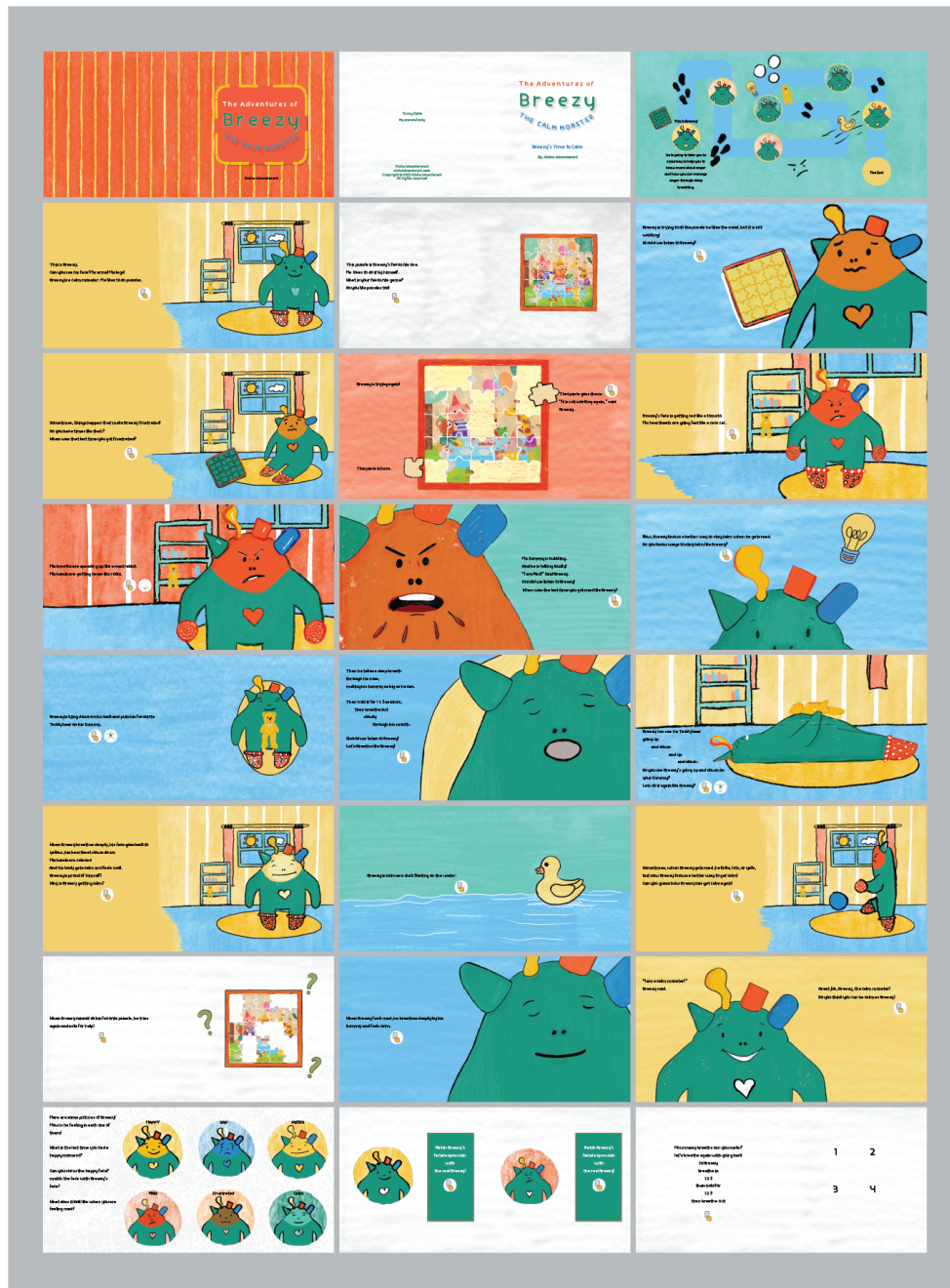


Figure 4: The final storybook prototype with illustrations

animated (app) directions effectively provide a complete modelling tool that allows parents to execute the ER skill. The interactive app integrated into the soft toy, which creates a TUI, enables parents to explain each emotion using digital facial expressions and various bodily expressions while interacting with the soft toy. By having three components (i.e., the soft toy, the app, and the storybook), the design creates a parent-child interaction that involves parents in teaching the strategy, provides them with a flexible experience to

use, and is suitable for them and their child's needs. Giving parents the freedom to select how they want to get involved in the play could enhance their active involvement and the implementation of teaching, which is necessary for young children. While playing with a physical stuffed toy, children can explore sensory experiences by investigating the different textures, colours, and sounds associated with the ears or body parts. Using interactive toys to facilitate a simple and practical ER technique could be an engaging way to



Figure 5: The belly deep breathing exercise with breezy

encourage parents to use this regulation strategy at home regularly. This playful interaction will not only influence ER learning and application in a home context, but it will also contribute to the positive development of parents' practises and young children's EC development. Overall, the soft toy is a useful and interactive tool for parents to use when teaching their children proper breathing techniques (Figure 5).

4 FUTURE WORK

In future work, we intend to make the interactive system applicable to all child populations. As a result, we will build on our findings and create an EC kit that will allow us to explore characters that explain various ER skills. Future research could also focus on the toy's efficacy in teaching ER strategies. Child development psychologists, parents, and cultural probes can be used to assess the toy's effectiveness. Furthermore, the long-term effects of using the toy prototype could be assessed by evaluating the use of the toy system several months after the initial intervention. Our long-term goals include conducting extended studies with parents and their children using a refined version of the toy design to understand the parent-child interaction with the intervention better.

5 CONCLUSION

By addressing the needs of early childhood anger dysregulation, which is associated with aggression and peer rejection, and attempting to support parents and young children at home, this research showcases a novel opportunity for playful interactions and pervasive technology design. This paper presents Breezy the calm monster: a soft toy that aims to support parents in teaching their young children how to identify and manage anger by modelling deep breathing. In this paper, the playful interaction needed the use of ubiquitous technology to offer interactive experiences that respond to the situation, the physical toy allowed children to create a relationship with the system without being exposed to technology constantly, and the storybook facilitated a familiar medium for parents to be involved in the experience. Parents offer emotional support to young children and play a significant role in teaching them ER skills. Tangible, interactive, and playful interactions have the potential to facilitate and improve parent-child interactions in everyday home environments.

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